

선박 사고 상황에서의 개인별 생존률 예측

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I. 탐구의 필요성 및 목적

1. 선박 사고의 위험성

- 선박사고로 인한 국내 사망자는 연간 160여명으로, 2900여명인 자동차 사고에 비해 피해 규모가 크진 않음.
- 하지만 선박사고는 한번 사고가 나면 대형 참사로 커지기 쉬움.

왜 선박 사고는 대형 참사로 커지기 쉬운가?



2. 개인별 생존률 예측의 중요성

- 선박 사고 시 생존률을 예측할 수 있다면, 생존률이 낮은 사람들을 유사시 우선적으로 대피시켜 총 사망자 수를 줄일 수 있음

II. 이론적 배경

- 개인별 생존률에 대한 데이터는 존재하나, 개인 별 생존률을 결정하는 factor는 거의 분석되지 않음.
- 다만, 사고 시 여성과 아이를 먼저 탈출시키는 "Woman and Children First" 정책이 있기에, 여성일수록/어릴수록 생존률이 더 높을것으로 예측되어옴

III. 탐구 방법 및 과정

- Elinder의 논문에서 지난 300년간 18개 선박사고의 15000명에 대한 데이터를 정리해 제공함

Ship	Data source	
	Source	Year
HMS Birkenhead	British Library, 1852. https://www.britishlibrary.org/reading-room/shipwreck	1852
SS Arctic	The New York Times, 1854. https://www.nytimes.com/1854/04/10/nyregion/shipwreck.html	1854
SS Golden Gate	The New York Times, 1852. https://www.nytimes.com/1852/04/10/nyregion/shipwreck.html	1852
SS Northern	Society for Promoting Christian Knowledge, 1873. https://www.scpk.org.uk/shipwreck	1873
RMS Atlantic	Cochran, C. Clark, R., 2000. https://www.cochran.org.uk/shipwreck	2000
SS Princess Alice	Princess Alice Rescue, 2011. Website: http://www.princessalice.org.uk/shipwreck	2011
SS Tampa	Salas, P.A., 2004. https://www.salas.org.uk/shipwreck	2004
RMS Titanic	Empire of France, 2004. Website: https://www.empireoffrance.org.uk/shipwreck	2004
RMS Empress of Ireland	Empire of Ireland Rescue, 2011. Website: https://www.empireofireland.org.uk/shipwreck	2011
RMS Lusitania	Lusitania Rescue, 2012. Website: https://www.lusitania.org.uk/shipwreck	2012
SS Princess Victoria	Princess Victoria Rescue, 2011. Website: https://www.princessvictoria.org.uk/shipwreck	2011
SS Vesta	The New York Times, 1928. https://www.nytimes.com/1928/04/10/nyregion/shipwreck.html	1928
SS Mary Celest	Empire of France, 2012. Website: https://www.empireoffrance.org.uk/shipwreck	2012
Admiral Nautilus	Admiral Nautilus Rescue, 2012. Website: https://www.admiralnautilus.org.uk/shipwreck	2012
SS Adriatic	SS Adriatic Rescue, 2011. Website: https://www.ssadriatic.org.uk/shipwreck	2011
SS Adriatic	SS Adriatic Rescue, 2011. Website: https://www.ssadriatic.org.uk/shipwreck	2011
SS Adriatic	SS Adriatic Rescue, 2011. Website: https://www.ssadriatic.org.uk/shipwreck	2011

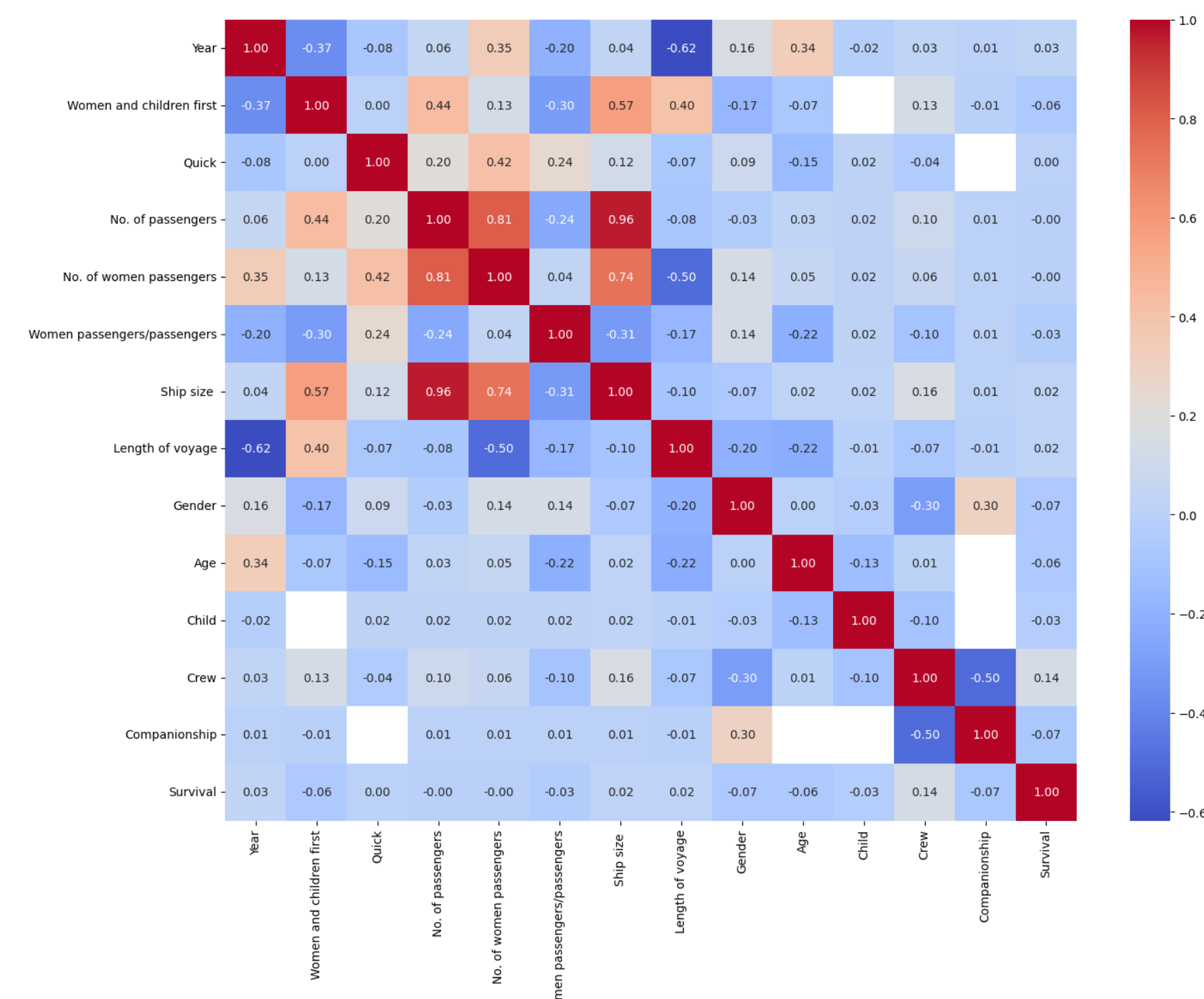
- Step 1: 해당 데이터를 아래의 코드로 로딩

```
import seaborn as sns
import pandas as pd
import matplotlib.pyplot as plt
train = pd.read_csv('선박 최종 데이터.csv')
train.head()
```

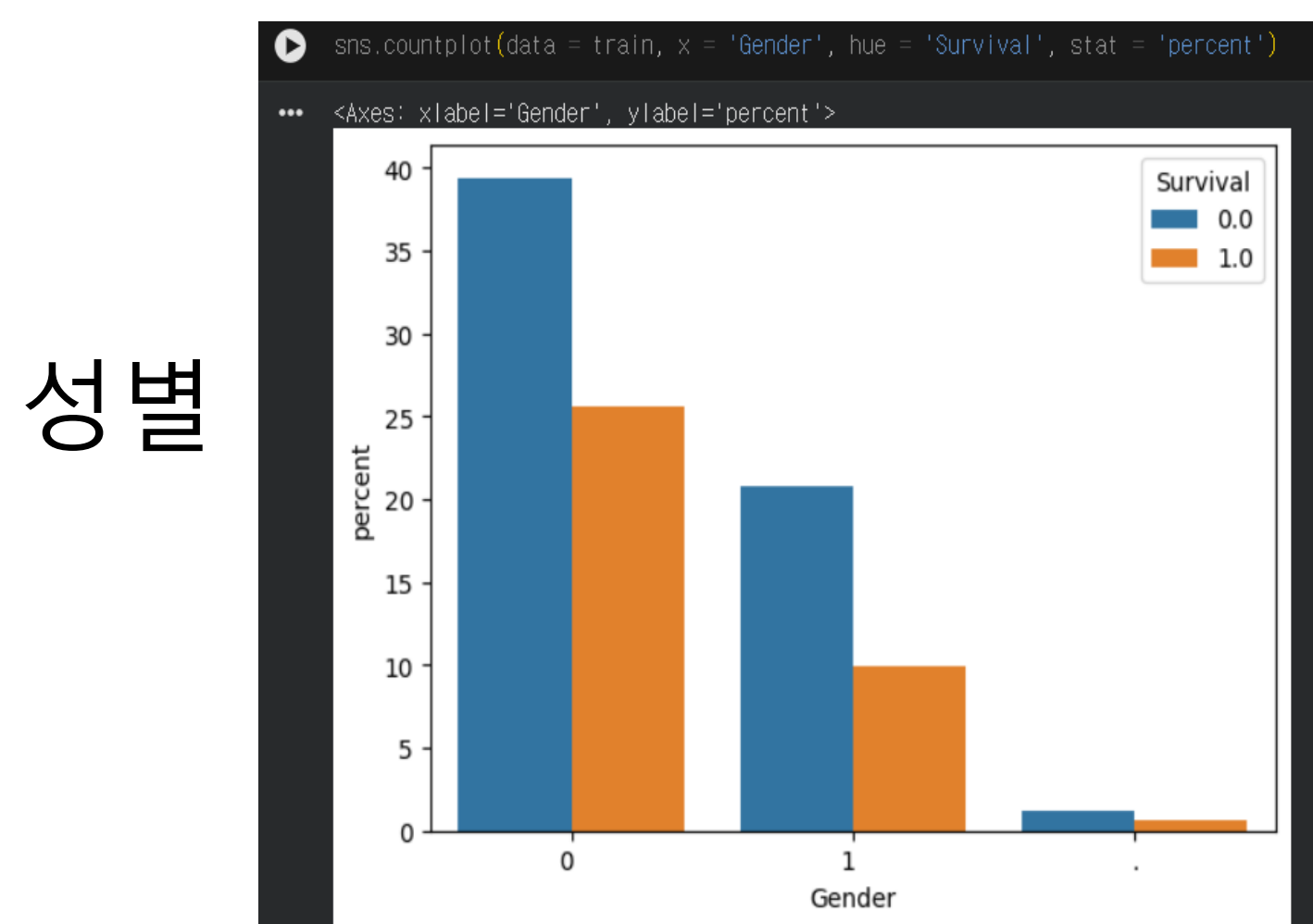
	Id_1	Ship Id	Year	Nationality of the Ship	Women and children first	Quick	Cause	No. of passengers	No. of women passengers/passengers	Women Ship size	Length of voyage	
0	1	HMS Birkenhead	1852	U.K	1	1	Grounding	490	7	0	554	21.0
1	2	HMS Birkenhead	1852	U.K	1	1	Grounding	490	7	0	554	21.0
2	3	HMS Birkenhead	1852	U.K	1	1	Grounding	490	7	0	554	21.0
3	4	HMS Birkenhead	1852	U.K	1	1	Grounding	490	7	0	554	21.0
4	5	HMS Birkenhead	1852	U.K	1	1	Grounding	490	7	0	554	21.0

- Step 2: 상관관계 매트릭스 생성

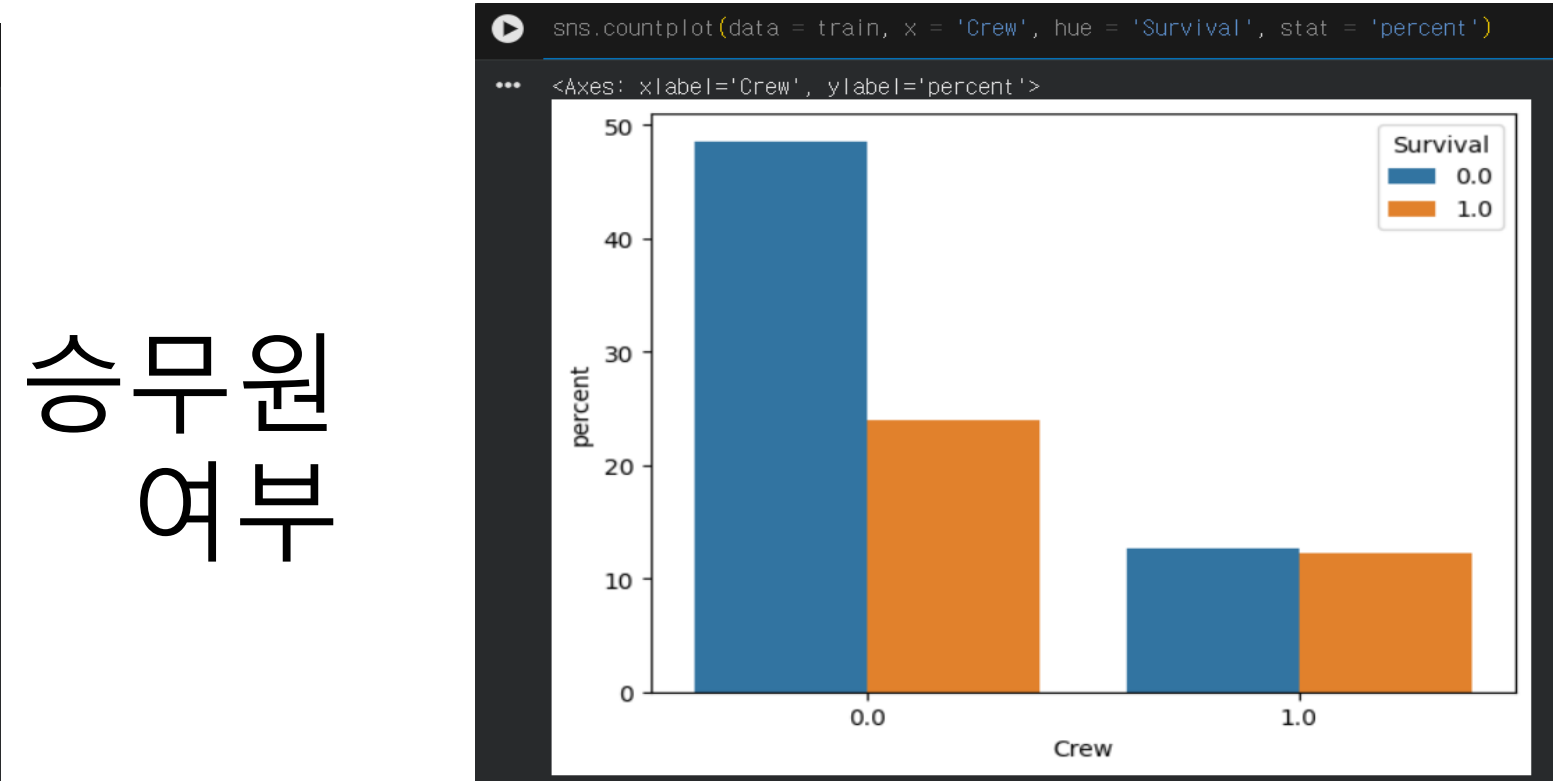
```
import seaborn as sns
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
train = pd.read_csv('선박 최종 데이터.csv')
cols = ['Gender', 'Age', 'Child', 'Crew', 'Companionship', 'Survival']
for col in cols:
    train[col] = pd.to_numeric(train[col], errors='coerce')
new_train = train.select_dtypes(include=[np.number])
train_corr = new_train.corr(method='pearson')
print(train_corr.columns)
plt.figure(figsize=(15, 12))
sns.heatmap(train_corr, annot=True, fmt='.2f', cmap='coolwarm')
plt.tight_layout()
plt.show()
```



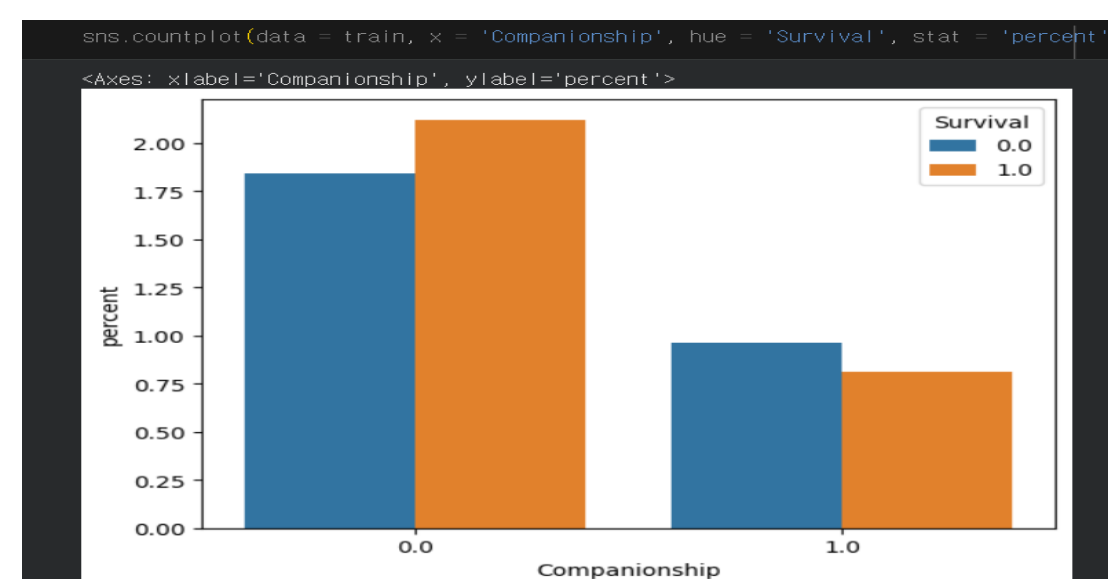
- Step 3: 생존률 결정 후보 factor들의 검증



성별

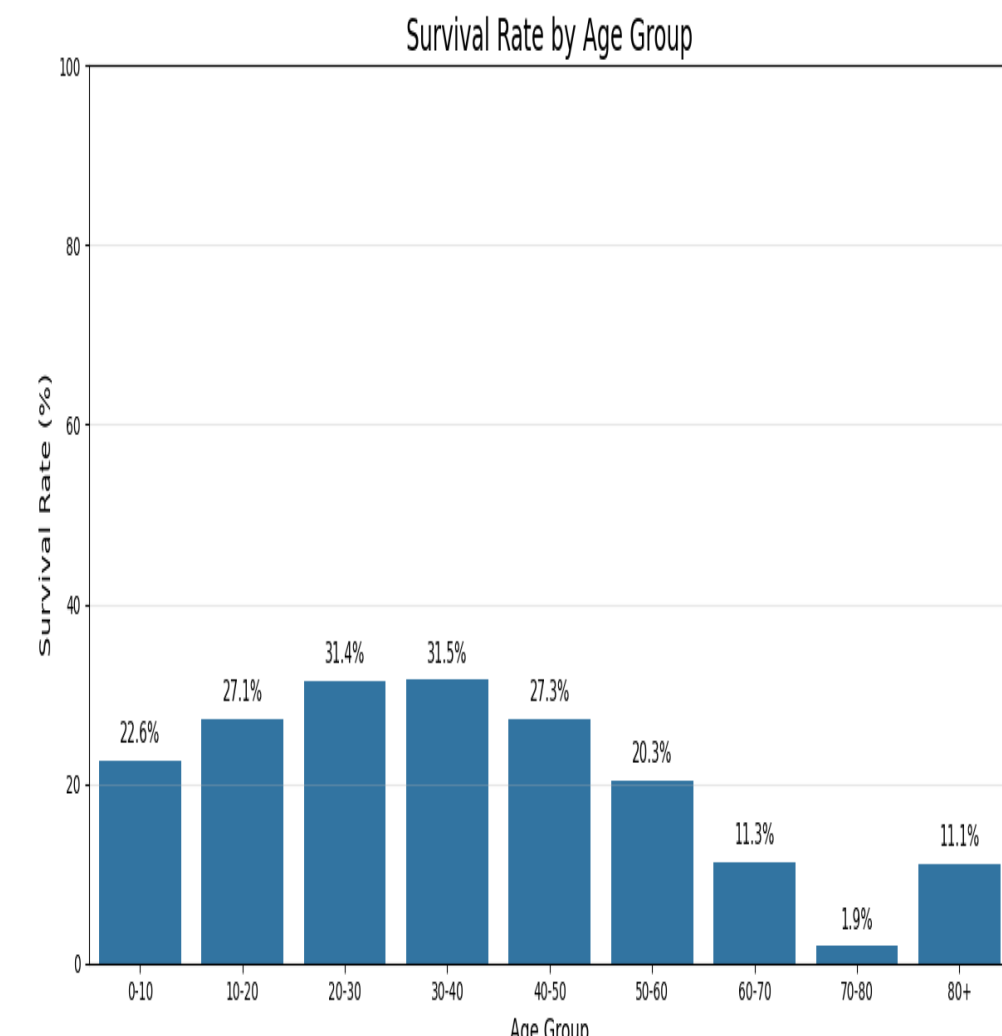


승무원 여부



동승객 여부

나이



- Step 4: 검증된 factor 들로 선박사고 시 생존 확률을 예측하는 로지스틱 회귀모델 제작

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

train = pd.read_csv('선박 최종 데이터.csv')
features = ['Gender', 'Age', 'Child', 'Crew', 'Companionship']
target = 'Survival'

train = train.reset_index(drop=True)
X = train[features]
y = train[target]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

model = LogisticRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print('모델 정확도: %.6f' % accuracy)
```

IV. 연구 결과

```
new_person = pd.DataFrame({
    'Gender': [0],
    'Age': [1],
    'Child': [0],
    'Crew': [1],
    'Companionship': [1]
})

new_person_scaled = scaler.transform(new_person)

probability = model.predict_proba(new_person_scaled)[0][1]

print(f'생존 확률: {probability * 100:.2f}%')
```

- 승객 정보로 생존확률을 예측하는 모델 완성

V. 결론 및 제언

- 개인별 생존률을 예측하는 모델의 개발 성공!

- 적용분야

- 선박이용시 사망보험금 산출
- 선박 내 자리 배치에 사용
- 선박사고시 구조순서 결정에 사용